

Boonyakorn ‘Than’ Thanpanit

I am an EECS student at MIT interested in embedded systems and FPGAs. In addition to formal coursework, I have more than a decade of experience as an electronics hobbyist. In high school, I won a gold medal at IOAA, a silver medal at IPhO, and a bronze medal at APPhO.

📍 Cambridge, MA ✉️ thanpa28@mit.edu ☎️ (617) 251-8612 💻 [than-thanpanit](#) 🔄 [polya2005](#)
 🔗 polya2005.github.io/portfolio/#

Education

BS **Massachusetts Institute of Technology**, Electrical Engineering with Computing Cambridge, MA
 • **Relevant coursework:** Computer Architecture, Digital System Lab (FPGA), Operating Systems, Low-Level Programming, Circuits, Control, Signal Processing, PCB Design
 Sept 2024 – present

Skills

Programming: C/C++, Python, RISC-V Assembly, C#, Web Development (HTML, CSS, Javascript, Typescript, Tailwind, React)

Hardware & Embedded: FPGA, Verilog, Bluespec, Vivado, Arduino, ESP-IDF, PlatformIO

CAD & Fabrication: KiCad, Fusion 360, 3D Printing (FDM/SLA), Laser Cutting

Software & Development Tools: Git, Linux, GNU Make

Other Technical Skills: LaTeX, Unity

Experience

MIT EECS, Circuits (6.200) Lab Assistant Feb 2026 – present
 • Confirmed the students’ understanding of key concepts in the labs of the Circuits course 8 months
 • Clarified class concepts to students during Lab Hours

MIT Metropolis Makerspace, Mentor Sept 2026 – present
 • Assisted shop users with tools such as 3D printers and laser cutters 1 month
 • Ran shop orientation for new members
 • Maintained the makerspace and the tools in the shop

MIT CSAIL, Undergraduate Researcher (UROP) Sept 2025 – May 2026
 • Reduced the latency of the Puppetmaster transaction scheduler by more than 90% 9 months
 • Improved the flexibility of the Bloom filter in the Puppetmaster transaction scheduler with Fibonacci hashing

MIT Game Developers Guild, Programmer Nov 2024 – Oct 2025
 • Implemented games in the Unity game engine 1 year
 • Operated in a team of writers, artists, musicians, programmers, and game designers
 • Collaborated both in person and online via Discord and GitHub

Projects

IoT RGB LED Matrix Jan 2026 – present
 • Developed a custom driver that translates SPI commands into control signals suitable for the LED matrix
 • Designed and assembled modular PCBs: 25 8x8 matrix modules and 5 interface modules
 • Developed the firmware for an ESP32-C6 module to interface with the IoT control and the matrix driver

FPGA Spinthesizer

Oct 2025 – Dec 2025

- Developed an RTL design that drives stepper motors to generate different pitches at different volumes according to the MIDI input
- Tested Verilog modules using Python and Cocotb
- Collaborated with Git in a team of two

Star Pointing and Identification Device

Mar 2021 – present

- Developed a device that automatically points its pointer toward the star selected by the user from more than 1000 stars in the built-in database and identifies the star that the user manually points the pointer toward
- Designed a PCB around the Raspberry Pi Pico, interfacing with peripherals via SPI, I2C, and UART interfaces
- Modeled the enclosure using Fusion 360 and fabricated it with 3D printing
- [GitHub repository for the first prototype](#) and [for the new, in progress version](#)